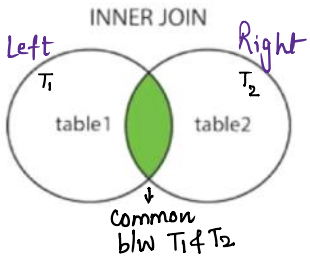
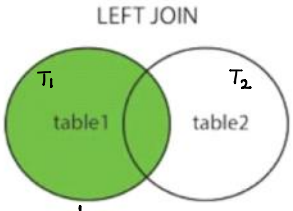
Combining DataFrames

* **Joins** :- To join / combine two tables.

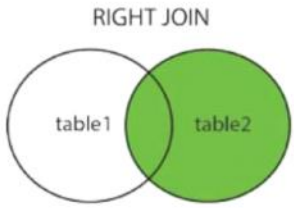
Ways to join tables :-

- Inner Join
- Left Join
- Right Join
- Full outer Join

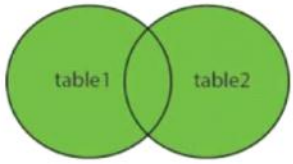




entire T_1 (left) & common from T_2



Common from and entire T_2 T_1



Entire T_1 & T_2

Inner Join :- Matches / combines only what is common b/w both the tables

T_1

	userid	name
0	1	jack
1	2	eliza
2	3	tom

Inner Join

T_2

	userid	msg
0	1	yes
1	1	okay
2	2	alright
3	4	nice

Merged Table

	userid	name	msg
0	1	jack	yes
1	1	jack	okay
2	2	eliza	alright

Left Join :- Takes entire Table T_1 (left) & only common / matching from Table T_2 (right)

T_1

Left

	userid	name

T_2

Right

	userid	msg
0	1	yes

Left

	userid	name
0	1	jack
1	2	eliza
2	3	tom

Right

	userid	msg
0	1	yes
1	1	okay
2	2	alright
3	4	nice

userid	name	msg
1	Jack	yes
1	Jack	okay
2	eliza	alright
3	tom	NaN

⟺

	userid	name	msg
0	1	jack	yes
1	1	jack	okay
2	2	eliza	alright
3	3	tom	NaN

Right Join :- Takes only common / matching rows from left Table T_1 & takes the entire Table T_2

T_1

LEFT

	userid	name
0	1	jack
1	2	eliza
~~2~~	~~3~~	~~tom~~

T_2

RIGHT

	userid	msg
~~0~~	1	yes
~~1~~	1	okay
~~2~~	2	alright
3	4	nice

userid	name	msg
1	Jack	yes
1	Jack	okay
2	Eliza	Alright
4	NaN	Nice

⟷

	userid	name	msg
0	1	jack	yes
1	1	jack	okay
2	2	eliza	alright
3	4	NaN	nice

Outer / Full Outer Join :- It merges the complete / entire rows of both the tables T_1 & T_2. [all matching & non-matching]

T_1

LEFT

	userid	name

T_2

RIGHT

	userid	msg

T_1 LEFT :

	userid	name
0	1	jack
1	2	eliza
2	3	tom

T_2 RIGHT :

	userid	msg
0	1	yes
1	1	okay
2	2	alright
3	4	nice

userid	name	msg
1	Jack	yes
1	Jack	okay
2	eliza	alright
3	Tom	NaN
4	NaN	nice

	userid	name	msg
0	1	jack	yes
1	1	jack	okay
2	2	eliza	alright
3	3	tom	NaN
4	4	NaN	nice

Ex- Table A & Table B

Id	Name
1	w
2	x
3	y
4	z

Id	Age
1	27
2	33
5	35
6	29

A.merge(B, on='Id', how='left') ⟶

1	w	27
2	x	33
3	y	NaN
4	z	NaN

B.merge(A, on='Id', how='right') ⟶

1	w	27
2	x	33
3	y	NaN
4	z	NaN